Chapter 1

Literature Review

1.1 Introduction

1.1.1 Scope and Objectives of the Review

Lumbar degenerative disease is among the most common indications for magnetic resonance imaging (MRI) worldwide, and the interpretation of those images is a repetitive, subjective and time-consuming task that has attracted sustained attention from the medical image analysis community [1, 2]. This chapter reviews that body of work. Its purpose is not to catalogue every published model, but to establish what is genuinely settled, what remains contested, and where a controlled experiment can still add knowledge.

The review is organised around a single guiding question: *what determines whether an automated system can grade lumbar degenerative severity reliably enough to be clinically useful?* Four candidate answers recur in the literature—the quality and definition of the reference labels, the degree to which the anatomy is localised before it is classified, the choice of representational backbone, and the configuration of imaging sequences supplied to the model. Each is examined in turn, and the chapter closes by identifying the two of these that remain inadequately resolved and that this thesis therefore addresses.

Three specific objectives follow from this. First, to characterise the clinical grading systems that supply the ground truth for every supervised model in this field, and to quantify the reliability ceiling those systems impose. Second, to trace the methodological evolution from handcrafted features to contemporary convolutional and attention-based architectures, with particular attention to comparative claims between the two. Third, to assess the evidence on how much diagnostic information is carried by each MRI sequence and plane, and thus whether the computational expense of multi-sequence processing is warranted.

1.1.2 Search Strategy, Sources, and Inclusion Criteria

The literature underpinning this chapter was assembled into a structured inventory of 92 records spanning 2001 to 2026. Sources were identified through PubMed, Scopus, IEEE Xplore and arXiv, using combinations of the terms *lumbar spine*, *MRI*, *spinal stenosis*, *disc degeneration*, *deep learning*, *convolutional neural network*, *transformer*, *segmentation* and *grading*. Reference lists of the retrieved systematic reviews [3–6] were searched by hand for records the database queries had missed, and the citation graphs of the most influential frameworks were traced forward.

Records were retained if they satisfied at least one of four criteria: they define or validate a clinical grading standard for lumbar degenerative disease; they present an automated method for detecting, segmenting or grading lumbar pathology on MRI; they externally validate such a method; or they describe a dataset or benchmark on which such methods are trained. Studies confined to computed tomography or plain radiography were excluded unless they establish a methodological principle transferable to MRI, as were studies of the cervical or thoracic spine except where explicitly used for contrast.

Two caveats on the evidence base should be recorded at the outset. First, the field's publication rate accelerated sharply after 2024, and a substantial fraction of the most directly relevant work [7–10] exists as preprints that have not completed peer review; these are cited as indicative of methodological direction rather than as settled results. Second, performance figures reported across studies are frequently not comparable, because they derive from different datasets, different label definitions and different evaluation protocols. Wherever numerical results are quoted in this chapter, the cohort and validation regime that produced them are stated alongside, and cross-study comparisons are avoided unless the underlying protocols genuinely align.

1.1.3 Structure of the Chapter

Sections 1.2 to 1.4 establish the clinical substrate: the disease, the imaging protocol used to observe it, and the grading systems that convert observation into ordinal labels. This material is not background decoration. The severity classes a model is trained to predict are defined entirely by these systems, and their documented unreliability propagates directly into every reported accuracy figure in the chapters that follow.

Section 1.5 traces the methodological history of the field, and Section 1.6 isolates anatomical localisation as the design principle that most consistently separates effective systems from ineffective ones. Sections 1.7 and 1.8 then address the two axes on which this thesis experiments: the representational backbone, and the spatial and sequential context supplied to it. Section 1.9 treats class imbalance and the ordinal structure of the label space, and Section 1.10 the datasets and benchmarks—principally the RSNA LumbarDISC collection—on which the field now converges. Sections 1.11 and 1.12 consider validation practice and interpretability respectively. Section 1.13 synthesises the whole and states the research gap.

1.2 Clinical Background: Lumbar Degenerative Disease

1.2.1 Epidemiology and Socioeconomic Burden

Low back pain is the single largest contributor to years lived with disability worldwide, and lumbar degenerative change is among its most frequently implicated structural correlates [3, 11]. Lifetime prevalence is reported as high as 84%, and in the United States alone the combined direct medical and indirect productivity costs are estimated between $50 and $100 billion annually [12]. Lumbar spinal stenosis in particular is the most common indication for spinal surgery in patients over 65 [13, 14], and demographic ageing is expected to increase both its incidence and the imaging workload it generates [1].

The operational consequence for this thesis is one of volume. MRI referrals for low back pain have grown faster than the radiology workforce able to report them [3, 15], and each lumbar study requires the reader to assign severity at multiple anatomical levels across several sequences—an average of 14 to 19 minutes of expert time per study [3]. It is this combination of high volume, repetitive structure and documented inter-reader variability that makes the task a plausible target for automation, rather than any claim that machines perceive the underlying pathology more acutely than radiologists do.

1.2.2 The Degenerative Cascade

Degeneration of the lumbar spine is a progressive, multifactorial process driven by ageing, mechanical loading, genetic predisposition and lifestyle factors [16, 17]. It begins in the intervertebral disc, whose nucleus pulposus loses proteoglycan content and therefore water from approximately the third decade of life onward. The resulting dehydration reduces the disc's height, its capacity to distribute axial load, and its signal intensity on T2-weighted imaging—the last of these being precisely what the Pfirrmann scale measures [18].

Disc height loss initiates a compensatory cascade through the rest of the motion segment. The annulus fibrosus weakens and may permit bulging or frank herniation of nuclear material; the ligamentum flavum buckles and hypertrophies; the posterior facet joints, now bearing altered load, develop osteoarthritic change and osteophytosis [14, 19]. Changes in the adjacent vertebral endplates and marrow—the Modic phenotypes [20]—frequently accompany this process and have themselves been the target of automated detection [21, 22]. The clinical significance of the cascade is that these individually modest changes act cumulatively to encroach on the spaces occupied by neural tissue.

1.2.3 The Three Anatomical Forms of Stenosis

Encroachment is conventionally partitioned by anatomical compartment, and this partition is directly mirrored in the label structure of the dataset used in this thesis [12]:

1. **Spinal canal stenosis** is narrowing of the central canal, which contains the thecal sac and the cauda equina. It is best appreciated on axial T2-weighted images, where the relationship between cerebrospinal fluid and nerve rootlets is visible [13, 23]. Severe central stenosis characteristically produces neurogenic claudication.

2. **Neural foraminal narrowing** is constriction of the lateral foramina through which the exiting nerve roots pass. It is assessed on parasagittal images, where the epidural fat surrounding the root provides the contrast that makes compression visible—T1 weighting being particularly suited to this [24]. It produces dermatomal radicular symptoms.

3. **Subarticular stenosis**, also termed lateral recess stenosis, affects the zone between the medial edge of the facet and the foramen, where it compresses the traversing root before it exits at the level below [25, 26].

The distinction matters computationally as well as clinically, because the three compartments are not equally visible in any single plane. Foraminal narrowing is poorly assessed from axial images and subarticular stenosis poorly from sagittal ones, a constraint that Section 1.8 shows to have concrete consequences for model design. It is also the compartment that correlates with which sequence a model must be given, and thus sits at the centre of the multi-sequence question this thesis examines.

1.2.4 Clinical Presentation and the Treatment Decision

Severity grading is not an academic exercise; it partitions patients into distinct management pathways. Conservative management—physiotherapy, analgesia, epidural steroid injection—is appropriate for mild and most moderate disease, whereas severe compression with corresponding symptoms may warrant decompressive surgery or fusion [13, 27]. The evidence that morphological grade carries genuine prognostic weight is strongest for the Schizas classification, where grade C and D patients were substantially more likely to fail conservative treatment [13], and for the MSU herniation grades, where size 2 and 3 lesions identify microdiscectomy candidates [27].

Two qualifications temper this. First, the correlation between imaging findings and symptoms is imperfect: anatomical stenosis is demonstrable in a substantial proportion of asymptomatic individuals [16, 25], so imaging severity constrains rather than determines management. Second, and more consequentially for automated systems, the clinically decisive boundary is not evenly distributed across the ordinal scale. The distinction between *normal* and *mild* rarely changes management, whereas the distinction between *moderate* and *severe* frequently does. A model that distributes its errors uniformly across grade boundaries is therefore not clinically equivalent to one of the same overall accuracy that concentrates its errors at the harmless end—a point developed in Section 1.9.6.

1.3 MRI Acquisition: Sequences, Planes, and Diagnostic Yield

1.3.1 Why MRI is the Reference Modality

MRI is the reference standard for non-invasive assessment of the lumbar spine. Computed tomography and plain radiography depict osseous anatomy well but resolve soft tissue poorly, and both expose the patient to ionising radiation. MRI resolves the intervertebral disc, thecal sac, individual nerve roots and ligamentous structures with a contrast unavailable to the alternatives [5, 17], and it is the disposition of exactly these soft-tissue structures that defines every grading system reviewed in Section 1.4. MRI has correspondingly displaced CT myelography for stenosis assessment [28].

1.3.2 Complementary Information Across Sequences and Planes

Lumbar protocols are inherently multiplanar and multisequence, because the spine is a three-dimensional structure whose pathologies are not all visible in one projection. Three acquisitions dominate clinical practice and constitute the input available in the dataset used here [12].

Sagittal T2-weighted imaging renders fluid hyperintense, so cerebrospinal fluid in the thecal sac appears bright against darker ligament and bone. It is the sequence on which disc hydration—and therefore Pfirrmann grade—is judged [18], and on which the craniocaudal extent of canal compromise is surveyed. Short tau inversion recovery (STIR) suppresses the signal from fat and thereby exposes marrow oedema and inflammatory change, making it the sequence of choice for active Modic type 1 change [21].

Sagittal T1-weighted imaging renders fat hyperintense. Because the exiting nerve root within the neural foramen is normally surrounded by epidural fat, T1 provides the contrast against which obliteration of that fat—the basis of the Lee foraminal grading system—becomes visible [24].

Axial T2-weighted imaging provides the transverse cross-section through the disc level. It is the only view in which the morphology of the dural sac, the arrangement of the cauda equina rootlets and the lateral recesses can be directly assessed, and it therefore underpins both the Schizas and Lee central canal systems [13, 23] and the assessment of subarticular stenosis [26].

1.3.3 Matching Pathology to Sequence

The correspondence between pathology and sequence is not a matter of convention but of physics, and the empirical literature bears this out with unusual clarity. The most direct evidence comes from external validation of sagittal-only systems. Nigru et al. evaluated SpineNetV2 across eleven radiological features on 1,747 lumbosacral discs and found agreement above $\kappa = 0.7$ for most targets but distinctly lower agreement for foraminal stenosis and disc herniation, attributing the shortfall specifically to the absence of transverse anatomical context in sagittal input [29]. Hallinan et al. approached the same constraint from the opposite direction, deliberately assigning axial T2 to central canal and lateral recess grading while reserving sagittal T1 for the neural foramina [30].

This has a direct bearing on the experiments in this thesis. If a single-sequence model is to be evaluated, the choice of which sequence to retain is not arbitrary: sagittal T2 is the most defensible candidate because it is acquired in essentially every back-pain protocol and because it carries the signal for canal stenosis and disc degeneration, but the literature predicts that it should be weakest precisely on the foraminal and subarticular targets. Whether a sufficiently expressive architecture can compensate for that predicted weakness is an open question, and one this thesis is designed to answer.

1.3.4 Sources of Acquisition Variability

Models trained on one institution's images frequently degrade on another's, and the proximate causes lie in acquisition. Scanner manufacturer and field strength (typically 1.5 versus 3.0 T), slice thickness and spacing, in-plane resolution, echo and repetition times, the presence or absence of fat suppression, and patient positioning all vary between sites [6, 31]. Axial acquisitions vary further in whether slices form a continuous stack or are angled to lie parallel to each disc level—a difference the LumbarDISC inclusion criteria explicitly accommodate [12], and one that any model consuming axial data must tolerate.

Sáenz Gamboa et al. treat this variability as the research problem in its own right rather than as a nuisance, evaluating segmentation across multiple centres and acquisition parameters and finding that ensembles of U-Net variants were required to maintain performance [31]. Batra et al. make the complementary design argument: rather than assuming uniform histograms and a fixed slice count per study, their pipeline is built to accept a variable number of axial and sagittal slices, on the grounds that this reflects the conditions under which clinical data actually arrives [7]. Variability is thus not merely a threat to validity but a specification constraint, and Section 1.11.3 returns to how effectively the field has met it.

1.4 Clinical Grading Systems as the Origin of Ground Truth

Every supervised model reviewed in this chapter learns to reproduce labels generated by a human applying one of the systems described below. The properties of those systems—their definitions, their granularity and above all their reliability—therefore bound what any model trained on them can achieve. This section treats them as the measurement instruments they are.

1.4.1 Disc Degeneration: Pfirrmann and its Modified Extension

The Pfirrmann classification, introduced in 2001, remains the most widely adopted scheme for intervertebral disc degeneration and is applied to sagittal T2-weighted images [18]. It assigns one of five ordinal grades on the basis of four morphological criteria: homogeneity of the nucleus pulposus, distinction between nucleus and annulus, T2 signal intensity as a proxy for hydration, and disc height. Grade I denotes a homogeneous, bright, normally proportioned disc; grade V a collapsed disc of uniformly low signal.

Its virtues are simplicity and universality; its limitations are well documented. It is a subjective visual judgement, it is insensitive to change over short intervals, and it captures only intrinsic disc state, saying nothing about Modic change, high-intensity zones or the downstream compression of neural structures [16, 17]. It also suffers a ceiling effect in elderly cohorts, where most discs present as grade IV or V and clinically meaningful variation is compressed into the top of the scale. Griffith et al. addressed this with an eight-level modified scale supported by a 24-image reference panel, validated on 260 discs in 52 subjects of mean age 73; intra-observer agreement was excellent (weighted κ 0.79–0.91) and inter-observer agreement substantial (0.65–0.67), with most disagreement confined to a single grade [32].

Esposito et al. place these two systems in a wider context that is worth recording: a scoping review of 569 studies identified *93* distinct grading systems for lumbar disc degeneration—63 subjective, 25 quantitative—of which the Pfirrmann method accounted for just over half of all reported use [16]. The apparent standardisation of the field is therefore partly illusory, and comparisons of automated grading accuracy across studies must confirm that the same instrument was being reproduced.

1.4.2 Central Canal Stenosis: Schizas and Lee

Two morphological systems have largely displaced simple dural sac cross-sectional area (DSCA) thresholds for central canal assessment, on the grounds that area measurements correlate poorly with symptoms and overlap substantially between symptomatic and asymptomatic individuals [13].

The **Schizas classification** grades the morphology of the dural sac on axial T2 images by the ratio of cerebrospinal fluid to nerve rootlets. Grades A and B retain visible CSF—in A the rootlets occupy part of the sac, in B they fill it but remain individually discernible, giving a characteristic grainy appearance. Grade A is itself divided into four sub-grades by the position and extent of the rootlets: A1, lying dorsally and occupying less than half the sac; A2, dorsal and in contact with the dura in a horseshoe configuration; A3, dorsal but occupying more than half the sac; and A4, lying centrally and occupying the majority of it [13]. Grades C and D show no discernible rootlets and no CSF signal, distinguished from one another by whether posterior epidural fat is preserved (C) or obliterated (D) [13]. This seven-level structure is worth noting against the three-class scheme used in this thesis: a benchmark that collapses severity to Normal/Mild, Moderate and Severe is discarding distinctions the clinical instrument makes, which places an upper bound on how finely any model trained on it can discriminate. Applied to 95 subjects across surgical, conservatively managed and low-back-pain groups, intra- and inter-observer agreement were substantial and moderate respectively ($\kappa = 0.65$ and 0.44). Critically, area measurement over-diagnosed stenosis in 35 patients and under-diagnosed it in 12 relative to the morphological grade, and grades C and D predicted failure of conservative management [13].

The **Lee central canal system** offers a simpler four-grade alternative based on the degree of separation of the cauda equina rootlets on axial T2: grade 0 no stenosis; grade 1 obliteration of the anterior CSF space with all rootlets still separated; grade 2 some rootlets aggregated; grade 3 none separated [23]. Assessed by four musculoskeletal radiologists across 81 patients, it achieved inter-reader ICC of 0.730–0.953 and intra-reader κ of 0.863–0.900, and grades tracked cross-sectional area and anteroposterior diameter at nearly all levels—notably better agreement than the Schizas system, at the cost of finer morphological detail.

1.4.3 Neural Foraminal Stenosis: Lee and Sartoretti

The **Lee foraminal system** grades parasagittal images by the pattern of perineural fat obliteration and the morphology of the nerve root: grade 0 normal; grade 1 fat obliteration in two opposing directions; grade 2 obliteration in all four directions without morphological change to the root; grade 3 nerve root collapse or distortion [24]. Evaluated on 576 foramina in 96 patients from L3–L4 to L5–S1 by two radiologists, inter- and intra-observer agreement were near-perfect (κ approximately 0.80–1.0). This is the direct clinical precursor of the left and right foraminal narrowing targets in the benchmark used in this thesis. The Sartoretti classification offers a six-point sagittal alternative (grades A–F) indexed to progressive contact between the exiting root and each foraminal boundary.

The near-perfect agreement Lee et al. report deserves scepticism when generalised. It was obtained by two experienced readers applying explicit criteria to a curated series; the multi-reader, multi-institution agreement figures in Section 1.4.6 are considerably lower, and it is the latter that describe the conditions under which large training corpora are actually annotated.

1.4.4 Subarticular and Lateral Recess Stenosis

Subarticular stenosis is graded on axial T2 by the proportion of the lateral recess compromised: mild at up to one third, moderate between one and two thirds, severe beyond two thirds, with nerve root involvement described as touching, displacing or compressing [25, 26]. Compared with the disc and central canal systems this is a notably less crisply specified instrument, resting on a visually estimated ratio rather than a discrete morphological pattern. That imprecision shows in the reliability data: subarticular stenosis is consistently the least reproducible of the targets [25], and it is also the target on which the severity distribution in LumbarDISC is least skewed (Section 1.10.4). Its difficulty is thus intrinsic, not merely a consequence of scarce severe examples.

Because the lateral recess is bounded posteriorly by the facet joint, facet degeneration contributes directly to subarticular narrowing, and it is graded by its own instrument—the four-point Fujiwara scale for facet joint osteoarthritis, assessed on T1 and T2 images by joint space narrowing, osteophyte formation, articular process hypertrophy and subchondral erosion. The Fujiwara scale is not among the RSNA targets, but it is instructive here because it represents the reliability floor of this family of instruments: Nikpasand et al. report average human agreement of only $\kappa = 0.13$ for Fujiwara grading, against 0.68 for Pfirrmann on the same cohort, and their CNN correspondingly reached $\kappa = 0.18$ against 0.68 [19]. Where the human instrument is that unreliable, no model trained on it can be expected to do better, and the authors read their own result as evidence that facet scoring needs better criteria rather than a better network.

1.4.5 Disc Herniation: Nomenclature and the MSU Classification

Herniation is described qualitatively by the standardised nomenclature of bulge, protrusion, extrusion and sequestration [33], and quantitatively by the Michigan State University (MSU) classification, which grades size against a transverse line joining the medial edges of the facet joints on the axial slice of maximal herniation [27]. Grade 1 extends less than half the distance to that line, grade 2 more than half without crossing it, grade 3 completely beyond it. The MSU system supplies the reference standard for several automated herniation studies [34], and its size 1 versus size 2/3 boundary maps onto the conservative versus surgical decision.

1.4.6 Inter- and Intra-Observer Reliability

The most consequential study for present purposes is that of Lurie et al., who assessed standardised MRI interpretation of lumbar stenosis using 58 randomly selected studies from the SPORT trial read by four independent readers, with 20 re-read after at least a month [25]. Inter-reader agreement was substantial for central stenosis ($\kappa = 0.73$) but only moderate for foraminal stenosis (0.58) and nerve root impingement (0.51), and poorest for subarticular stenosis (0.49), with marked variation between reader pairs. Quantitative measurement fared no better in absolute terms: mean absolute inter-reader difference in thecal sac area was 128 mm^2, or 13%. Intra-reader agreement exceeded inter-reader agreement for every feature.

Two structural findings recur across the wider reliability literature. First, agreement is systematically worse for the lateral compartments than for the central canal—the same ordering that later appears in automated model performance. Second, disagreement is overwhelmingly confined to adjacent grades: Griffith et al. report 84% of intra-observer and 91% of inter-observer disagreement spanning a single grade [32], and the same pattern recurs in automated evaluation [35, 36]. Earlier consensus efforts confirm that the underlying definitions were themselves contested: the Delphi survey of Mamisch et al. [37] and the consensus conference reported by Andreisek et al. [26] were both convened precisely because radiologic criteria for stenosis varied between institutions.

1.4.7 Label Noise as a Performance Ceiling

The reliability figures above are not a peripheral caveat; they are a hard constraint on the interpretation of every result in this field. If four expert readers agree on subarticular severity at $\kappa = 0.49$, then a model trained on one reader's labels and evaluated against another's cannot exceed that agreement, and a reported accuracy approaching unity on such a target should prompt scrutiny of the evaluation protocol rather than admiration. The LumbarDISC dataset description states the point explicitly, noting that published inter-rater agreement on these grades ranges from poor to substantial depending on anatomical location [12].

The field has responded to this in two ways. The first is to make the reference standard more robust by construction—majority voting across a panel with adjudication of disagreements, as in Tumko et al. (seven radiologists) [28] and Su et al. (two clinicians with expert adjudication) [38]. The second is to reframe the evaluation question: rather than asking whether a model matches a single reader, asking whether its disagreement with readers falls inside the band of disagreement between readers. Won et al. furnish the cleanest example, reporting that the difference between their two experts was statistically significant while the difference between each expert and the model trained on that expert's labels was not [39]. Comparatively few studies adopt either safeguard, and Section 1.11.7

returns to this as a systemic weakness.

1.4.8 The Ordinal Structure of the Label Space

Severity labels are ordinal, not categorical: *normal*, *mild*, *moderate* and *severe* stand in a fixed order, and the distance between them is neither uniform nor clinically symmetric. Treating them as unordered classes under categorical cross-entropy—as the majority of the reviewed work does—discards that structure, penalising a severe-graded-as-normal error exactly as heavily as a moderate-graded-as-severe one.

Two empirical observations follow. First, errors concentrate overwhelmingly at adjacent grade boundaries. McSweeney et al. report that SpineNet disagreed with human raters on 20.83% of discs but that only 0.85% of discs differed by more than one grade [35]; Udomluck et al. observe the same concentration [36]; Ishimoto et al. find exact-grade agreement of 65.7% rising to 94.1% when collapsed to a severe-versus-rest dichotomy [40]. Much of the apparent error in ordinal grading is therefore boundary error rather than gross misclassification.

Second, and less expectedly, explicitly ordinal objectives have not reliably improved on this. Niemeyer et al. tested soft-kappa loss, ordinal cross-entropy and regression losses against plain cross-entropy for Pfirrmann grading on 1,599 patients and 7,948 discs, together with class pooling and class-weighted losses, and found that none improved overall classification performance; the default cross-entropy classifier reached $\kappa = 0.92$ with predictions within one grade of ground truth in 99.2% of cases [41]. Patil et al. likewise found performance strongest at the extremes of the Pfirrmann scale and on the binary low-versus-high distinction, with intermediate grades remaining the difficulty [42]. The ordinal structure of the problem is thus real and measurable, but the question of how to exploit it is genuinely open—a point that informs the loss functions adopted in Chapter 3.

1.5 Evolution of Automated Lumbar MRI Analysis

1.5.1 Handcrafted Features and Classical Machine Learning

Before deep learning, automated analysis of lumbar MRI depended on features designed by hand. Investigators specified mathematical descriptors of local appearance and passed them to a conventional classifier. Ghosh et al. combined Histogram of Oriented Gradients (HOG) descriptors with a support vector machine and a set of geometric heuristics, reporting 99% disc localisation accuracy across 53 clinical cases and 318 discs, and—unusually for the period—emitting a tight bounding box for each disc rather than a single interior point, so that a downstream system could operate on the region directly [43]. Athertya and Kumar took a comparable approach to vertebral degeneration, extracting local binary patterns, Hu moments and gray-level co-occurrence matrix (GLCM) statistics to classify Modic changes, endplate defects and focal changes, reaching 85.91% accuracy for the extent of Modic change on 100 patients [22]. Related efforts applied hybrid classical models to disc abnormality detection [44, 45] and semi-automatic computer-aided classification of degenerative disease [46], while Huber et al. pursued texture analysis with decision trees as a quantitative alternative to visual stenosis grading [47]. He et al. applied a synchronised superpixel representation to define consistent regions of interest across levels before classifying neural foraminal stenosis, an early attack on one of the exact conditions graded in this thesis [48].

These systems established feasibility but generalised poorly. Handcrafted descriptors are sensitive to scanner hardware, field strength, slice thickness and patient positioning, and the anatomical variability of the lumbar spine defeats fixed geometric assumptions. One finding from this era nevertheless retains value: Athertya and Kumar's feature-sensitivity analysis showed that GLCM entropy alone sufficed to detect focal changes, while endplate-defect classification depended almost entirely on the first two Hu moments [22]. Such explicit attribution of predictive power to individual features is something the deep models that displaced these methods recovered only later, and imperfectly, through the interpretability techniques of Section 1.12.

1.5.2 The First End-to-End Deep Learning Systems

Convolutional neural networks removed the need for explicit feature design by learning hierarchical representations directly from pixels [49]. In lumbar imaging the decisive contribution was SpineNet [50]. Trained on 2,009 patients and 12,018 disc volumes from sagittal T2 images, it predicted six radiological scores simultaneously—Pfirrmann grade, disc narrowing, upper and lower endplate defects, and upper and lower marrow changes—from a shared convolutional trunk that branched into task-specific heads. Two design decisions proved durable. First, multi-task learning let one architecture serve several targets, reducing the free parameters relative to training separate networks and exploiting the correlation between co-occurring pathologies. Second, the system required only disc-level class labels: no voxel-wise segmentation and no slice-level localisation, with the evidence hotspots emerging implicitly from classification training. Reported performance was state-of-the-art and near-human across all six scores.

The same group demonstrated shortly afterwards that longitudinal clinical scans could serve as free supervision. A Siamese network was pre-trained with a contrastive loss on whether two scans belonged to the same patient—information available from DICOM metadata without knowing the patient's identity—together with an auxiliary vertebral-level prediction task, then fine-tuned for degeneration grading. On 1,016 subjects, 423 with follow-up imaging, the pre-trained network outperformed training from scratch and reached equivalent performance from far fewer annotated examples [51]. This anticipated by several years the self-supervised pretraining strategies now routine in medical imaging, and it is the direct ancestor of the contrastive approach discussed in Section 1.9.5.

SpineNet's principal limitation was structural rather than architectural: it consumed preprocessed sagittal disc volumes, and could therefore not assess pathologies that require the transverse plane. SpineNetV2 extended coverage to eleven radiological features and moved to a multi-stage design, using a U-Net to detect and label vertebral bodies and discs before a ResNet classifier graded each [52]. The reliance on sagittal input nonetheless persisted, and Section 1.11.3 shows this to be the consistent locus of its external-validation failures.

1.5.3 From Binary Detection to Multi-Level, Multi-Pathology Grading

The task definition itself has evolved, and the trajectory matters more than any individual model. Early deep learning work asked binary questions: is stenosis present or absent [53]; is a herniation visible in this image [54]. Such formulations are tractable but clinically thin, because management depends on severity and on which level is affected.

Three shifts followed. The first was to *multi-class severity*: Won et al. graded stenosis status on 542 L4–5 axial images against two independent experts [39]; Niemeyer et al. predicted the full five-point Pfirrmann scale across 7,948 discs [41]. The second was to *multi-pathology* output, in which one model reports several findings at once: Lehnen et al. evaluated disc herniation, bulging, canal stenosis, nerve-root compression and spondylolisthesis across 888 lumbar segments from a single CNN [55]; Su et al. graded herniation, central canal stenosis and nerve-root compression from a shared ResNet-50 trunk with three classification heads [38]. The third was to *per-level, per-side* prediction, in which the unit of analysis is not the study or the image but the individual disc level and, for the lateral compartments, the individual side. Hallinan et al. exemplify the mature form, grading central canal, lateral recess and neural foraminal stenosis at each level with the sequence appropriate to each [30]. Single-stage detectors have since been adapted to deliver the same output more cheaply: Wang et al. modified a YOLOv5 network to detect regions of interest and grade central, lateral recess and foraminal stenosis simultaneously [56], and a related design grades Pfirrmann degeneration, herniation and high-intensity zones from sagittal and axial input in one pass [57]. Minin et al. pursued the same objective for Pfirrmann grading with a YOLOv8x model released openly, motivated by the inaccessibility of proprietary graders [58]. The progression through the YOLO generations—v3 for herniation detection [54], v5 for stenosis and degeneration grading [56, 57, 59], v8 for combined segmentation and keypoint localisation [58, 60]—tracks the field's gradual shift from detecting a lesion to characterising it.

This third formulation is precisely that of the benchmark used in this thesis, which requires twenty-five graded outputs per study: five conditions across five disc levels [12, 61]. The progression is therefore not merely historical. It explains why performance figures from the binary era cannot be compared with contemporary results, and why a model that classifies a whole image is solving a different and substantially easier problem than one that must attribute severity to a specific anatomical location.

1.5.4 Chronological Synthesis

Four phases are discernible. Until roughly 2016 the field was dominated by handcrafted features and classical classifiers, with localisation as the central problem [15, 43]. From 2017 to 2020, CNNs displaced these methods and multi-task grading became feasible [50, 62, 63]. From 2021 to 2023 the emphasis moved to clinical credibility: external validation [35], interpretability [14, 21], reader-assistance evaluation [64] and multi-centre testing [38, 65]. From 2024 onward, the release of a large public benchmark [12] redirected effort toward anatomically localised, multi-view severity grading, alongside the arrival of transformer and vision-language architectures [7, 9, 10, 66].

The through-line is a steady increase in the specificity of what is being predicted, accompanied—until recently—by a decrease in the rigour with which it is validated. Both systematic reviews of the period make the same complaint: Compte et al. found that algorithms performed markedly worse in replication or external validation than in their development studies [3], and Wang et al. concluded that despite pooled sensitivity of 0.84 and specificity of 0.87 across twelve studies and 15,044 patients, no model was reliable enough for clinical practice [4]. Section 1.11 takes up this tension directly.

1.6 Anatomical Localization as a Design Principle

1.6.1 Disc and Vertebra Detection, Labelling, and Level Assignment

Before severity can be attributed to L4–L5, the system must know which structure is L4–L5. Level assignment is thus a precondition of per-level grading, and it has been treated as a problem in its own right since the earliest work [43, 67]. Pan et al. illustrate the mature solution: a Faster R-CNN locates vertebral bodies L1–S1 on sagittal images, the geometric relationship between sagittal and axial acquisitions identifies the corresponding disc in each axial slice, and the disc region of interest is then extracted—each of the three stages reported at 100% accuracy, with the residual difficulty falling entirely on the subsequent classification [68]. Detection has, in short, largely been solved; grading has not.

1.6.2 Semantic and Instance Segmentation of Lumbar Structures

Dense segmentation provides a stronger anatomical substrate than bounding boxes. The U-Net architecture [69] and its descendants dominate, as a dedicated systematic review of deep-learning-assisted disc segmentation confirms [70]. Ghosh and Chaudhary segmented vertebrae, disc, dural sac and background jointly using cascaded random forests over image and sparsely sampled context features, achieving Dice 0.87 and 0.84 for dural sac and disc across 212 clinical cases [15]. Han et al. introduced adversarial training for multi-structure segmentation [63], Sáenz-Gamboa et al. established CNN semantic segmentation of the structural elements surrounding the spinal cord [71], and Al-Kafri et al. applied SegNet to boundary delineation for stenosis assessment across 515 symptomatic patients, contributing two purpose-built metrics—confidence and consistency—for assessing the quality of the ground truth itself rather than only the model [72].

Subsequent work has largely consisted of architectural refinement. Wang et al. added multilevel feature-map attention and residual blocks to an Attention U-Net [73]; Ahmed et al. targeted class imbalance between anatomical structures with a custom combined loss and modified upsampling [74]; Silveira et al. added an Inception module for multi-scale extraction and a dual-output mechanism, reporting mIoU 0.8974 and F1 0.9444 on SPIDER and—notably—outperforming TransUNet [75]. Möller et al. extended coverage to the whole spine with SPINEPS, segmenting fourteen structures including the posterior elements, and established a useful general result: segmenting semantically first and instance-wise second outperformed training directly on instance segmentation, beating an nnU-Net baseline [76] on every structure and metric [77]. Chen et al. merged convolutional and attention pathways in parallel with a novel position embedding, outperforming fifteen competing models [78].

1.6.3 Public Segmentation Resources and Benchmarks

Progress in this subfield has been unusually well served by shared data. The SPIDER dataset provides 447 sagittal T1 and T2 series from 218 patients across four hospitals, with reference segmentations of vertebrae, discs and spinal canal plus per-level radiological gradings, together with reference performance for a baseline algorithm and nnU-Net and a continuous public challenge on a sequestered test split [79]. Several of the segmentation models above are trained and benchmarked on it [42, 74, 75], which makes their results mutually comparable in a way that is rare elsewhere in this literature.

1.6.4 Quantitative Morphometry

Segmentation enables measurement, and measurement offers an escape from ordinal subjectivity. The dural sac cross-sectional area is the canonical example. Ghobrial and Roth compared U-Net, Attention U-Net and MultiResUNet for automated DSCA quantification on 683 scans with 50 held back for external validation; MultiResUNet performed best, with Pearson correlation 0.9917 and mean absolute error 23.70 mm^2, holding up externally at 99.95% accuracy and F1 0.9393 [80]. Zheng et al. pursued the same objective for disc degeneration, segmenting degeneration-related regions with a Swin-transformer network and computing signal-intensity and geometric features, then proposing a quantitative criterion to replace subjective Pfirrmann grading on the basis of strong correlation with grade across a large population [11]. Related work measures the lordosis angle from segmentation masks [81].

A caution is warranted. Ghobrial and Roth's model relies on T1 weighting alone and a modest dataset, as the authors acknowledge [80], and Schizas' original finding—that area measurement over- and under-diagnosed stenosis relative to morphological grade in a substantial minority of patients [13]—applies equally whether the area is traced by hand or by network. Automating a measurement does not repair a weak correlation between that measurement and the clinical state.

1.6.5 Evidence that Localisation Precedes Classification

The strongest recurring methodological finding in this literature is that explicit anatomical localisation before classification improves grading. The argument was made structurally by DeepSPINE, which chained U-Net vertebral segmentation and disc-level designation to a multi-task grading network across 4,075 patients and more than 22,000 disc-level annotations [62], and it has been corroborated repeatedly since.

Four lines of evidence converge. Bharadwaj et al. segmented dural sac and disc with V-Net, localised facet and foramen by geometric rule, and only then classified, reaching Dice 0.93 and 0.94 with foramen and facet localisation IoU 0.72 and 0.83 [14]. Šušteršič et al. showed that segmenting, cropping and enhancing the region of interest before classification produced 0.87 and 0.91 accuracy on axial and sagittal views for a multi-class herniation problem [82]. Kowlagi et al. made the comparative case most directly, arguing that a well-tuned three-stage pipeline of segmentation, localisation and classification allows convolutional networks to outperform transformer approaches on Pfirrmann grading in a population cohort [83]. Acharya and Kansakar tested the principle in its strongest form, hypothesising that anatomical focus rather than architecture is the binding constraint, and grading severity from a single localised region of interest per disc [8].

The implication for the present work is direct. If localisation dominates architecture in determining performance, then a comparison between backbones is only meaningful when both receive identically localised input—otherwise the comparison measures preprocessing rather than representation. This is a methodological requirement that Section 1.7.5 shows much of the existing comparative literature fails to satisfy.

1.6.6 Reduced-Annotation and Unsupervised Alternatives

Dense annotation is expensive, and several strands of work reduce the requirement. Kuang et al. eliminated manual labels entirely for vertebral segmentation: sub-optimal masks from a rule-based method were combined through a voting mechanism to supervise CNN training, with a regional strategy restricting attention to a specific vertebral region, validated across 243 volunteers [84]. SpineNet's evidence hotspots represent the complementary weakly supervised route, obtaining localisation from classification labels alone [50], while the SPIDER annotation protocol adopted a human-in-the-loop compromise, training on a small subset to produce initial masks that were then corrected and fed back [79]. Self-supervised pretraining on unlabelled or metadata-labelled data [51] completes the picture. Together these establish that the annotation burden, long cited as the principal obstacle to clinical deployment, is substantially negotiable.

1.7 Backbone Architectures for Severity Classification

1.7.1 Convolutional Networks

Convolutional networks exploit two properties well matched to imaging: local connectivity, since a filter responds to a neighbourhood rather than the whole field; and translation equivariance, since the same filter is applied at every position. Stacking convolutions builds a hierarchy in which early layers respond to edges and texture and later layers to composite anatomical structure.

ResNet addressed the degradation encountered when networks are made deeper, introducing residual blocks whose skip connections allow a layer's input to bypass subsequent convolutions and be added to their output, providing an unimpeded path for gradients and permitting stable training at fifty layers and beyond [49]. In lumbar imaging ResNet is ubiquitous: as the classification head in multi-task grading [38], as the detection backbone in staged pipelines [34], as the encoder in segmentation networks [7], and as the classifier in SpineNetV2 [52].

EfficientNet attacked a different problem—how to allocate a fixed computational budget—by scaling depth, width and input resolution together under fixed coefficients derived from a neural architecture search, rather than scaling one dimension in isolation [85]. The resulting models reach comparable accuracy at substantially lower parameter and FLOP cost, which matters for the deployment scenarios that motivate this thesis. EfficientNet appears as the classifier in Kowlagi et al.'s Pfirrmann pipeline [83] and in the final stage of M-SCAN [7]. **ConvNeXt** represents the most recent convolutional line, modernising the CNN with design choices borrowed from transformers while retaining convolution [86]; Udomluck et al. include ConvNeXt-Tiny in their feature extractor comparison and find it stable across classifiers [36].

The architectural limitation of convolution is a bounded receptive field. Deep stacks aggregate local responses into progressively broader ones, but the network never explicitly models a relationship between two distant regions—for instance, between the L1–L2 and L5–S1 levels, whose degenerative states are biomechanically coupled.

1.7.2 Vision Transformers and the Attention Bottleneck

Transformers dispense with convolution and rely on self-attention, in which every element of the input is weighted against every other regardless of distance [87]. The Vision Transformer applies this by partitioning an image into fixed patches, flattening them and treating the result as a sequence [88]. Global context is available from the first layer, and there is no locality prior to overcome—but also no locality prior to exploit, which is why vision transformers are markedly more data-hungry than convolutional networks of comparable capacity.

The practical obstacle is cost. Global self-attention scales quadratically with the number of patches, hence quadratically with image area. Clinical MRI demands high spatial resolution to resolve structures a few millimetres across, so a naive vision transformer over full-resolution lumbar images is computationally prohibitive.

1.7.3 The Swin Transformer

The Swin Transformer resolves this bottleneck through two mechanisms [89]. *Windowed self-attention* restricts attention to non-overlapping local windows, making cost linear rather than quadratic in image size; to restore cross-window communication, the window partition is *shifted* by half a window between consecutive layers, so information propagates across boundaries over successive blocks. *Hierarchical feature maps* are produced by patch-merging layers that progressively reduce spatial resolution while increasing channel depth, yielding a pyramid analogous to a CNN's and making the backbone usable for dense prediction as well as classification.

For lumbar grading the appeal is specific. Severity depends simultaneously on fine local detail—the texture of a nucleus pulposus, the obliteration of perineural fat—and on regional context, since a rootlet configuration is interpretable only relative to the surrounding dural sac and the levels above and below. A hierarchical, windowed attention mechanism is a plausible fit to that structure, which is the substantive reason for selecting it as the transformer representative in this thesis. Swin-derived architectures have been applied to spinal segmentation, and Zheng et al. incorporate Swin components in their quantitative degeneration network [11].

1.7.4 Hybrid CNN–Transformer Architectures

A third position holds that the two families are complementary. SymTC merges convolutional and transformer layers in a *parallel dual-path* arrangement rather than stacking them sequentially, and integrates a position embedding into the self-attention module to improve spatial awareness; benchmarked against fifteen existing models on both a private dataset and a released synthetic one, it performed best for both vertebral bones and discs [78]. Zheng et al. similarly combine convolutional and Swin components [11], and Yi et al. pair a 3D ResNet18 backbone with transformer components across three hospitals [65]. Baur et al. explore a different pairing, segmenting discs with a U-Net, reconstructing each as a 3D point cloud and grading it with a graph neural network; performance was uneven across levels—F1 0.71 at L2/3 against 0.46 overall—which the authors present as a foundation requiring refinement rather than a competitive result [90]. Hybrids complicate the clean comparison this thesis undertakes, but they indicate where the field expects the answer to lie: not in the wholesale replacement of convolution, but in supplying it with a mechanism for long-range interaction.

1.7.5 Published CNN-versus-Transformer Comparisons and Their Limits

Several studies bear directly on the architectural question, and their results do not favour transformers as strongly as the general computer vision literature might suggest.

Kowlagi et al. address it explicitly. Observing that recent gains had been attributed to transformer architectures, they show that a well-tuned three-stage convolutional pipeline outperforms the state-of-the-art transformer approaches on Pfirrmann grading, evaluated on the Northern Finland Birth Cohort 1966—a population cohort rather than a clinical convenience sample—with an ablation study and subgroup breakdown, and public code [83]. Their conclusion is that pipeline design and tuning outweigh backbone choice.

Lee et al. provide the closest published analogue to the present study, developing a CNN-based and a transformer-based model for lumbar stenosis classification on 564 MRIs with a 100-study external test set, labelled by four radiologists with expert consensus as reference and read by eight participants across four experience levels. Both models exceeded 94% detection recall; on dichotomised classification the CNN, transformer and human participants achieved kappas of 0.99/0.99/0.97–0.98 for central canal, 0.98/0.94/0.81–0.94 for lateral recesses and 0.98/0.95/0.91–0.95 for neural foramina. The CNN was superior overall, the transformer similar to superior against clinicians [91].

Udomluck et al. compared VGG19, ConvNeXt-Tiny and DINOv2 features with classical classifiers under external validation, finding conventional VGG19 features strongest (accuracy 0.9556 with logistic regression) and DINOv2 features generalising worst, dropping to 0.6222 with LightGBM [36]. Patil et al. reached a still more pointed conclusion through ablation: radiomic features carried essentially the entire signal for Pfirrmann grading, and frozen ImageNet-pretrained ResNet50 features added no measurable discriminative value [42]. Silveira et al. likewise found their enhanced U-Net outperformed TransUNet on segmentation [75].

Three methodological limitations recur across this evidence. First, the compared architectures are rarely given identical preprocessing, localisation and training budgets, so differences may reflect tuning effort rather than representational capacity—exactly the confound Kowlagi et al. identify [83]. Second, the input configuration is usually held fixed, leaving unexamined the interaction between architecture and available context, even though a transformer's advantage should plausibly be largest where global context must substitute for missing views. Third, several comparisons rely on frozen pretrained features rather than fine-tuned backbones [36, 42], which disadvantages transformers disproportionately given their weaker inductive biases. These three gaps motivate the experimental design of this thesis.

1.7.6 Transfer Learning, Pretraining, and Self-Supervision

Medical datasets are small by the standards these architectures were designed for, so initialisation matters. ImageNet pretraining is near-universal, though Patil et al.'s ablation is a caution that its benefit is not automatic [42]. Domain-specific pretraining is better supported: Jamaludin et al. obtained equivalent performance from far fewer labels through longitudinal self-supervision [51], and Acharya and Kansakar used contrastive pretraining on disc-centred crops to reduce sensitivity to irrelevant appearance variation [8, 92]. Sequential transfer between related targets—initialising a subarticular model from spinal canal weights—has also been reported to help on smaller datasets, a strategy of particular relevance to transformers given their lack of convolutional inductive bias.

1.8 Exploiting Spatial Context: Slices, Volumes, and Views

1.8.1 2D, 2.5D, and 3D Input Representations

An MRI series is a volume, but severity is assigned per level, so a representation must be chosen. *2D* models classify individual slices and are cheap but discard through-plane context; since a disc spans several slices and a radiologist integrates across them, a single slice may not contain the evidence. *3D* models consume the volume directly [65] at considerable memory cost—Acharya and Kansakar note that a standard high-resolution series contains millions of voxels, making full-volume processing impractical for many architectures [8].

2.5D representations occupy the middle ground and have become the default in this domain. A small stack of adjacent slices is treated as channels or as a short sequence, capturing local through-plane context at roughly 2D cost. SpineNet's input was a stack of nine slices per disc [50]; M-SCAN describes its approach explicitly as 2.5D, selecting the three axial slices closest to each level and modelling them with recurrent layers [7]; the leading challenge solutions combined 2D CNNs with sequence models or attention-based multiple-instance learning over the slice dimension.

1.8.2 Cross-Sequence and Cross-Plane Fusion

Where multiple sequences are available, they must be combined. The simplest approach concatenates them as channels, which presumes spatial registration that sagittal and axial acquisitions do not satisfy. More considered strategies process each sequence separately and fuse at the feature level.

CrossSpine is the most explicit treatment. Nguyen et al. identify two failings of baselines—single-stream designs cannot integrate complementary information across sequences, and uniform treatment of all discs ignores level-specific anatomy—and address the first with a cross-sequence attention mechanism that fuses features at multiple spatial scales, letting each sequence attend to all others, followed by a weighting mechanism that prioritises the most diagnostically informative sequence. They report Macro F1 improved by more than 125%, Mean AUPRC by 99% and Mean AUROC by 36% over their baseline [9]. M-SCAN applies cross-attention to align sagittal and axial feature maps before classification [7], and Park et al. pursue transformer-based multimodal fusion on the same benchmark [93]. Zeng et al. extend fusion beyond imaging entirely, combining ConvNeXt image features with GPT-4o-generated textual descriptions encoded by RoBERTa through a Mamba fusion stage [66].

1.8.3 Geometric Correspondence Between Planes

Fusing sagittal and axial data requires knowing which axial slice corresponds to which sagittal level—a geometric problem solved through DICOM spatial metadata rather than learned. Pan et al. derive the correspondence from the geometric relationship between the two acquisitions, reporting 100% accuracy in identifying the disc in each axial slice [68]. M-SCAN projects 2D sagittal stenosis coordinates into 3D using patient orientation data from the DICOM headers, then selects the three nearest axial slices for each of the five levels [7].

This step is a hidden cost of multi-sequence modelling, and one that bears directly on the trade-off examined here. It adds a preprocessing dependency, a failure mode when metadata is absent or inconsistent, and an engineering burden that a single-sequence pipeline simply does not incur.

1.8.4 Multi-Sequence versus Single-Sequence Input

The clinical standard is unambiguous that all three sequences contribute [12, 30], and the strongest reported results on the RSNA benchmark use all three [7]. The countervailing consideration is cost: multi-sequence processing demands greater memory and compute, cross-plane registration, and longer inference, and the resulting infrastructure requirements exceed what many hospital environments provide.

The evidence on whether reduced input suffices is genuinely mixed, which is why the question remains open. On one side, external validation of sagittal-only SpineNetV2 found agreement dropping specifically for foraminal stenosis and herniation—the pathologies requiring transverse context [29]—and Wu et al. found ordinal Pfirrmann grading degrading in upper lumbar discs while binary detection held up [94]. On the other, Acharya and Kansakar grade severity from *sagittal T2 alone*, reaching 78.1% balanced accuracy with a severe-to-normal misclassification rate of 2.13%, on the argument that anatomical focus rather than input breadth is the binding constraint [8]. Kowlagi et al. likewise operate on sagittal input and outperform transformer baselines [83].

Two observations sharpen the question. First, the comparison has not been made under controlled conditions: the sagittal-only and multi-sequence results above come from different architectures, datasets, localisation strategies and evaluation protocols, so the contribution of sequence configuration cannot be isolated. Second, the expected penalty is target-dependent—severe for foraminal and subarticular grading, mild for central canal—so a single aggregate accuracy figure would obscure precisely the structure of interest. Both observations inform the design adopted in Chapter 3.

1.8.5 Robustness to Missing or Incomplete Sequences

Clinical data is incomplete: sequences are omitted, truncated or corrupted by artifact. The LumbarDISC inclusion criteria required all three sequences and excluded studies with substantial artifact [12], so a model trained on it has not been exposed to the degraded conditions of routine practice. A model that requires three sequences fails outright when one is absent, whereas a single-sequence model is unaffected—an argument for reduced input configurations that is about robustness rather than efficiency, and one that the literature on missing-modality learning addresses only in passing for this application.

1.9 Class Imbalance and Ordinal Severity Modelling

1.9.1 Severity Distributions in Lumbar MRI Datasets

Class imbalance in this domain is not an artefact of sampling but a property of the population: in any unselected clinical cohort, most disc levels are normal or mildly degenerate, and severe compression is uncommon. The RSNA LumbarDISC distributions make the scale explicit [12]. Of 13,474 spinal canal stenosis grades, 85.4% are normal or mild, 8.8% moderate and 5.9% severe. Of 26,919 neural foraminal grades, roughly 78% are normal or mild, 18% moderate and 4.5% severe. Subarticular grading is the least skewed of the three, with roughly 69% normal or mild, 19% moderate and 11% severe across 26,285 grades.

The same pattern recurs elsewhere. Liawrungrueang et al. report a Pfirrmann distribution of 220 grade I, 530 grade II, 170 grade III, 160 grade IV and just 20 grade V across 1,000 images [59]; Niemeyer et al. describe an approximately Gaussian grade distribution with grades 1 and 5 under-represented [41]. Two consequences follow. First, a model trained under unweighted cross-entropy can achieve roughly 85% accuracy on spinal canal stenosis by predicting *normal/mild* unconditionally, so aggregate accuracy is close to uninformative on this task. Second, the ordering of imbalance across targets is the inverse of the ordering of difficulty: subarticular stenosis has the most balanced label distribution yet the poorest human reliability [25], confirming that its difficulty is intrinsic rather than a consequence of scarce positive examples.

1.9.2 Data-Level Strategies

The most common data-level intervention is weighted random sampling, in which the probability of drawing a minority-class example is inflated so that each minibatch approaches a uniform severity distribution and gradients are not dominated by the majority class. Aggressive augmentation of minority classes complements this, since with few severe examples a network may memorise them rather than learn the morphology of severe compression. Reported techniques fall into three groups: intensity transformations, of which contrast-limited adaptive histogram equalisation (CLAHE) is the most common in this domain because it enhances local soft-tissue contrast without amplifying global noise; geometric transformations such as random scaling, affine rotation, horizontal flipping and elastic deformation; and sample-mixing methods such as MixUp and CutMix, which synthesise new training examples by blending or splicing pairs of images and their labels. Mixing methods warrant caution on ordinal targets, since interpolating between a normal and a severe disc produces a label with no clinical referent.

Tsai et al. provide the clearest single demonstration of augmentation's value in the small-data regime, treating it as the subject of the study rather than an implementation detail: using YOLOv3 with a DarkNet-53 backbone to detect disc herniation, mean average precision reached 92.4% at 550 images with augmentation, and augmentation was decisive in preventing both under- and over-fitting [54]. Ahmed et al. attack a related form of imbalance—between anatomical structures rather than severity classes—through preprocessing that rectifies class inconsistencies before training [74].

1.9.3 Loss-Level Strategies

Class-weighted cross-entropy multiplies each class's contribution to the loss by a scalar, forcing the optimiser to penalise minority-class errors more heavily. The RSNA challenge encoded this directly into its evaluation metric, assigning weights of 1, 2 and 4 to normal/mild, moderate and severe respectively, so that aligning the training objective with the evaluation criterion is a matter of adopting the same coefficients.

Focal loss goes further, introducing a modulating factor that down-weights well-classified examples dynamically and concentrates gradient on hard, borderline cases [95]. This is well matched to the present problem, where the abundant normal cases are mostly easy and the difficulty concentrates at the mild/moderate and moderate/severe boundaries—precisely where human readers also disagree. Acharya and Kansakar combine weighted focal loss with contrastive pretraining [8], and Islam Sk et al. propose an adaptive PID-Tversky loss that borrows feedback control to adjust training penalties dynamically toward under-segmented minority instances [10].

1.9.4 Ordinal-Aware Objectives and Whether They Help

Given the ordinal label structure of Section 1.4.8, it is natural to expect ordinal-aware objectives to outperform categorical ones. The evidence does not support this expectation as strongly as the intuition suggests.

Niemeyer et al. conducted the most systematic test available, evaluating soft kappa loss, ordinal cross-entropy and regression losses against standard cross-entropy for Pfirrmann grading on 1,599 patients and 7,948 discs, with extensive hyperparameter optimisation, and additionally trying class pooling and class-weighted losses to counter imbalance. None of these improved classification performance overall. The default cross-entropy classifier reached Cohen $\kappa = 0.92$, average sensitivity 90.2% and precision 92.5%, with predictions within one grade of ground truth in 99.2% of cases and a mean absolute error of 0.08 grades [41]. Patil et al. list ordinal regression among the supplementary analyses they ran, and still report that intermediate-grade discrimination remains the outstanding difficulty [42].

Two readings are possible. Either the ordinal structure is already implicitly captured—a network trained with cross-entropy on visually ordered classes will naturally confuse adjacent grades more often than distant ones—or the specific formulations tested were inadequate. The literature does not currently distinguish these, which is why this thesis treats the choice of objective as an empirical question rather than a settled one.

1.9.5 Contrastive and Representation-Learning Approaches

A third strategy attacks imbalance in the representation rather than in the sampling or the loss. Contrastive learning pulls examples of similar severity together in the latent space and pushes dissimilar ones apart before any classifier is fitted [92], so the representation is shaped by the structure of the problem rather than by class frequency.

Acharya and Kansakar apply this directly, combining contrastive pretraining with disc-level fine-tuning over a single anatomically localised region of interest per disc, on the reasoning that this forces attention onto intrinsic disc features and reduces sensitivity to irrelevant appearance variation. An auxiliary regression task handles disc localisation and a weighted focal loss addresses the residual imbalance. The result—78.1% balanced accuracy with the severe-to-normal misclassification rate reduced to 2.13% relative to supervised training from scratch—is notable less for the headline accuracy than for what it optimises [8]. The lineage runs back to Jamaludin et al.'s longitudinal self-supervision [51].

1.9.6 The Clinical Asymmetry of Error

The most important point in this section is that not all errors are equal, and that standard metrics conceal the difference. Grading a severe stenosis as normal risks a missed diagnosis and delayed decompression; grading a normal disc as moderate risks an unnecessary review. Overall accuracy, macro-F1 and even Cohen's κ treat these identically.

Three studies take the asymmetry seriously. Acharya and Kansakar report the severe-to-normal misclassification rate as a headline result alongside balanced accuracy, arguing explicitly that conventional metrics do not capture the differential clinical risk [8]. Wu et al. characterise SpineNetv2's error profile as specificity-oriented, with false negatives exceeding false positives, and conclude that the system is suitable as a confirmatory second reader but that reliance on its negative findings is not recommended—a conclusion about deployment that follows from the shape of the errors rather than their number [94]. The RSNA weighted metric encodes the same judgement into the benchmark itself.

Ishimoto et al. supply the complementary observation: collapsing four grades to a severe-versus-rest dichotomy raised agreement from 65.7% to 94.1% with $\kappa = 0.75$ [40]. If the clinically decisive question is whether a level is severe, then performance on that question is considerably better than the multi-class figures suggest, and evaluation should report both. This chapter therefore recommends, and Chapter 3 adopts, reporting per-class recall for the severe class alongside aggregate measures.

1.10 Datasets and Benchmarks

1.10.1 Private and Single-Institution Cohorts

Most work before 2024 used private single-institution data, with cohort sizes spanning three orders of magnitude—from 75 exams [21] and 146 consecutive patients [55] through 200 studies [14] and 542 images [39] to 4,075 patients [62] and 15,254 axial images [38]. Aggregate performance correlates only loosely with cohort size, because larger private datasets are frequently drawn from a single scanner and protocol.

Two consequences follow, and both are structural rather than incidental. Results are not comparable across studies, since each is measured against a different label distribution and reference standard; and models cannot be independently re-evaluated, since the data cannot be shared. Both systematic reviews of the period identify this as the field's central methodological weakness [3, 6].

1.10.2 Population and Epidemiological Cohorts

A smaller body of work uses population cohorts, which differ from clinical convenience samples in a way that matters for generalisation: participants are not selected by referral, so the severity distribution reflects the population rather than the imaged-because-symptomatic subset. The Northern Finland Birth Cohort 1966, comprising lumbar MRI from participants imaged at age 46–47, is the most used [35, 83]; the Wakayama Spine Study [40] and the Hong Kong Disc Degeneration Population-Based Cohort [96] serve similar roles. Kowlagi et al. are explicit that evaluating on a population cohort rather than a clinical sample is a deliberate methodological choice, reflecting the transferability of results to downstream research [83].

1.10.3 Public Lumbar MRI Datasets

Public data arrived late to this field relative to CT, where large annotated vertebral collections have existed for years. SPIDER is the principal segmentation resource: 447 sagittal T1 and T2 series from 218 patients across four hospitals, with reference segmentations of vertebrae, discs and spinal canal, per-level radiological gradings, baseline and nnU-Net reference performance, and a continuous challenge on a sequestered split [79]. Its influence is visible in the number of subsequent methods benchmarked on it [42, 74, 75, 80]. Smaller public collections support classification work [19, 97, 98].

1.10.4 The RSNA 2024 Challenge and LumbarDISC

The RSNA Lumbar Degenerative Imaging Spine Classification dataset is the largest and most diverse expertly annotated lumbar spondylosis collection assembled, and it is the dataset used in this thesis [12]. It comprises MRI studies from 2,697 patients and 8,593 image series drawn from eight institutions across six countries and five continents—a geographic spread chosen deliberately so that models trained on it face genuine acquisition heterogeneity rather than a single site's conventions.

Inclusion required a sagittal T2-like series (conventional spin echo T2, STIR or Dixon), a sagittal T1 series and an axial T2 series covering at least three lumbar disc levels, from outpatients aged 18 or over imaged for degenerative disease. Studies with lumbosacral hardware, active non-degenerative pathology, severe scoliosis or substantial artifact were excluded [12]. Annotation was performed by expert volunteer neuroradiologists and musculoskeletal radiologists recruited through the ASNR [61]. The associated challenge [61] made the dataset the field's common benchmark and shifted attention decisively from whole-image classification toward anatomically localised, per-level severity grading.

1.10.5 Label Structure, Distribution, and the Weighted Metric

The label structure is the defining feature of the benchmark and warrants precise statement. Each study receives twenty-five graded outputs: five conditions —spinal canal stenosis, left and right neural foraminal narrowing, left and right subarticular stenosis—at each of five disc levels from L1/L2 to L5/S1. Each is assigned one of three ordinal classes: Normal/Mild, Moderate or Severe. Expert coordinate annotations accompany the grades, so anatomical localisation can be learned in a supervised manner rather than inferred [12, 61].

Three properties follow. The task is *not* generic three-class image classification, and results from studies that treat it as such are not comparable. The severity distribution is severely imbalanced and differently so for each condition (Section 1.9.1). And the challenge metric is a sample-weighted log loss with weights of 1, 2 and 4 across the three classes, which encodes the clinical asymmetry of Section 1.9.6 directly into the evaluation. A model tuned for unweighted accuracy is therefore optimising a different objective from the one the benchmark measures.

1.10.6 Challenge Solutions and the Practices They Standardised

The solutions converged on a common template, and that convergence is itself a finding. Nearly all adopted multi-stage pipelines in which localisation precedes classification: a detection model—commonly a U-Net, YOLO variant or Faster R-CNN with a feature pyramid—identifies each disc level on the sagittal series, and a second-stage classifier grades tightly cropped regions, typically using 2.5D representations that combine a 2D backbone with a sequence model or attention over adjacent slices.

M-SCAN exemplifies the approach and reports AUROC 0.971 on 1,975 studies, using a U-Net with ResNet50 encoder to locate stenosis points on sagittal images, projecting these into 3D via DICOM orientation metadata to select corresponding axial slices, and applying cross-attention to align sagittal and axial features [7]. Liu et al. pair MobileNetV2 with U-Net for efficiency, reporting 94.93% accuracy under five-fold cross-validation [99]; Park et al. apply transformer-based multimodal fusion [93]; and further challenge-derived work continues to appear [100, 101].

Two cautions attach to these figures. Most are cross-validation results on the public training split rather than external validation, so they measure fit to LumbarDISC rather than transportability beyond it. And the top solutions rely on large ensembles, which inflate reported accuracy relative to what a single deployable model achieves—a gap directly relevant to the efficiency question this thesis examines.

1.11 Evaluation, Validation, and Generalization

1.11.1 Metrics and Their Appropriateness

Reported metrics in this field are heterogeneous and frequently ill-matched to the task. Overall accuracy remains the most common headline figure despite being close to meaningless under the severity distributions of Section 1.9.1. Area under the ROC curve is widely reported [4, 7] but is defined for binary problems and must be extended by one-versus-rest averaging for three-class grading, which conceals which class is failing.

More appropriate choices appear in the better studies. Cohen's and Gwet's κ measure agreement above chance and are directly comparable with the human reliability figures of Section 1.4.6, which is the correct reference point [35, 38, 64]. Balanced accuracy corrects for prevalence [8, 35]. Mean absolute error in grades exploits the ordinal structure and distinguishes a one-grade slip from a two-grade error [41, 94]. Ishimoto et al. report both exact-grade agreement and the collapsed severe-versus-rest figure, making the difference visible rather than choosing between them [40].

The recommendation that follows is to report a small panel rather than a single number: a prevalence-corrected aggregate, an agreement statistic comparable to inter-reader values, per-class recall for the severe class, and the distribution of error magnitudes across grade boundaries.

1.11.2 Internal Validation and Optimistic Reporting

A recurring pattern is that internally validated performance is high and externally validated performance is materially lower. Several reported accuracies in the reviewed literature exceed 95% [59, 98, 99, 102], which sits uneasily beside inter-reader agreement of $\kappa = 0.49$ for subarticular stenosis [25]. Where a model appears to exceed the reproducibility of the process that generated its labels, the explanation usually lies in the evaluation protocol:

single-institution data, cross-validation rather than held-out external testing, labels from a single reader, or a task collapsed to a binary decision.

The point is not that these studies are wrong on their own terms, but that their numbers do not transfer. Compte et al. state the general finding directly: algorithms did not perform as well in replication or external validation as in their development studies [3].

1.11.3 External Validation and Domain Shift

External validation on geographically and temporally separate cohorts is the strongest available evidence of clinical viability, and a small number of studies supply it.

McSweeney et al. evaluated SpineNet on 1,331 participants from the Northern Finland Birth Cohort 1966, a dataset separate from its UK training data in both geography and time. Balanced accuracy was 78% for disc degeneration and 86% for Modic changes, with Pfirrmann reliability against expert radiologists of Lin concordance 0.86 and Cohen $\kappa = 0.68$, and Modic detection at $\kappa = 0.74$; both were essentially unchanged in a low-back-pain subset. The informative detail is the error structure: 20.83% of discs were graded differently from human raters, but only 0.85% differed by more than one grade [35]. Nigru et al. evaluated SpineNetV2 across eleven features on 1,747 discs from 353 patients, finding agreement above 0.7 for most targets but distinctly lower for foraminal stenosis and herniation, attributed to the limits of sagittal input [29]. Wu et al. tested SpineNetv2 on 491 patients and 2,455 discs, reporting overall accuracy of 83.5–97.5% and superiority over a junior orthopaedic surgeon on canal stenosis, spondylolisthesis and bilateral foraminal stenosis [94]. Grob et al. contribute a further independent validation on 882 scans [103].

Studies that build external testing into their development are still uncommon but instructive. Tumko et al. evaluated on 150 studies graded by a panel of seven radiologists with majority voting and external adjudication [28]; Su et al. used a 1,273-image external set from a different hospital and observed accuracy falling by roughly 7–10 points relative to internal testing [38]; Zhang et al. found classification accuracy dropping from 87.70% to 74.23% externally [34]; Zeng et al. report an unusually small external drop, from 93.7% to 92.2% [66].

1.11.4 Detection Versus Fine-Grained Grading

A consistent and under-appreciated finding is that these two capabilities generalise differently. Zhang et al. provide the clearest illustration: moving to an external institution, detection degraded only modestly—mean IoU 0.82 to 0.70, precision 98.4% to 96.3%, sensitivity 99.4% to 97.8%—while classification accuracy fell from 87.70% to 74.23% [34]. Wu et al. report the same asymmetry in different terms: binary detection across five pathologies held up well while ordinal Pfirrmann grading remained the limiting factor, degrading further in older patients and upper lumbar discs [94]. Yilihamu et al. show the effect within a single study as task granularity increases, using a dual-branch YOLOv8 pipeline that pairs YOLOv8-seg for disc segmentation with YOLOv8-pose for canal keypoint localisation: segmentation IoU remained above 97% externally, while classification precision fell from 81.21% to 74.50% for an 18-category problem yet held at 92.51% to 90.07% for four-category severity [60].

The implication is that where a system fails is predictable. Finding the anatomy is robust; judging how bad it looks is not. Reporting a single aggregate figure for a pipeline that performs both obscures this, and evaluation should separate the two stages.

1.11.5 Benchmarking Against Clinicians

Comparison against human readers is the most clinically meaningful benchmark, and the better studies construct it carefully. Won et al. provide the cleanest design: two experts each labelled the data, a model was trained on each expert's labels, and the key result was that the difference between the two experts was statistically significant while the difference between each expert and their corresponding model was not [39]. Tumko et al. compared against a seven-radiologist panel average and found the model matched or exceeded it on all three stenosis types [28]. Lee et al. benchmarked against eight participants spanning four experience levels [91], and Wu et al. against orthopaedic surgeons of differing seniority [94].

Two cautions apply. Comparisons against junior readers or against surgeons rather than radiologists set a lower bar than subspecialty radiological practice [94, 104]. And readers in these studies typically work from cropped images without clinical history, which is not how they read in practice—so such comparisons measure a narrower task than clinical interpretation.

1.11.6 Longitudinal Stability and Reproducibility

One-shot accuracy does not establish that a model is stable over repeat imaging, which matters for monitoring progression. Murto et al. supply the only longitudinal evidence in this collection, comparing SpineNet's Pfirrmann grading against radiologists in 19 volunteers imaged at age 37 and again at 51. Radiologist-to-SpineNet concordance was 0.54 at both timepoints, clearly below the 0.83 and 0.78 observed between radiologists, and SpineNet systematically assigned grade 1 to several discs and grade 5 to more discs than the readers did [105]. The cohort is small, but a systematic offset that is stable across time is exactly the failure mode that one-shot evaluation cannot detect. Cheung et al. approach the temporal dimension differently, predicting five-year progression from baseline imaging across 1,343 participants [96].

1.11.7 Reporting Quality and Recurring Weaknesses

The systematic reviews converge on a consistent diagnosis. Wang et al. pooled twelve studies covering 15,044 patients, obtaining pooled sensitivity 0.84 and specificity 0.87 with SROC AUC 0.92, but with heterogeneity of I^2 near 99% and four of twelve studies at high risk of bias; they concluded that no current model is reliable enough for clinical practice, citing the need for accepted reference standards, larger datasets, external validation and fuller reporting [4]. Compte et al. found no significant difference between algorithm families and identified the development-to-validation performance drop as the field's central problem [3]. Mendes et al. found that most of 56 segmentation studies relied on private datasets and lacked external validation [6]. Liawrungrueang et al. report metrics spanning 71.5% to 99% across twenty studies—a range so wide as to indicate that the underlying tasks are not comparable [5].

Four weaknesses recur: single-institution data without external testing; ground truth from a single reader with no adjudication; task definitions that vary between studies while sharing a label vocabulary; and metric selection that flatters imbalanced problems. The present work adopts the public LumbarDISC benchmark, its panel-derived labels and its weighted metric specifically to avoid the first, second and fourth of these.

1.12 Interpretability and Clinical Integration

1.12.1 Saliency, Evidence Hotspots, and Grad-CAM

The earliest interpretability contribution in this domain is SpineNet's evidence hotspots, which map output-layer gradients back to the input to produce a heatmap per radiological score, obtained without any localisation supervision [50]. Grad-CAM [106] has since become the default. Tabarestani et al. use it to confirm that their cascade attends to clinically relevant regions [102], and Silveira et al. report that attention concentrated on vertebral bodies and nucleus pulposus in spatial correspondence with manual segmentations [75].

The value of these methods should not be overstated. A saliency map shows where a network responded, not why, and confirming that attention falls on plausible anatomy is a weak check—it excludes gross reliance on background artefact but does not establish that the model has learned the morphological criteria the grading system specifies.

1.12.2 Inherently Interpretable Models

A stronger position is to build models whose reasoning is legible by construction. Bharadwaj et al. provide the most striking result in this literature: alongside a Big Transfer classifier they trained a decision-tree classifier operating on segmentation-derived features for central canal stenosis, and the interpretable model *outperformed* the black box, reaching $\kappa = 0.80$ against 0.54 and approaching inter-reader agreement of 0.74–0.86 [14]. Quantitative morphometry belongs to the same family: a measured dural sac area [80] or a quantitative degeneration criterion [11] is inspectable in a way a logit is not. Patil et al. make the related point that radiomic features carried the entire discriminative signal for Pfirrmann grading while deep features added nothing measurable, and that radiomics additionally yields interpretable shape and intensity descriptors usable as imaging biomarkers [42].

Together these findings undercut the assumption that interpretability must be purchased at the cost of accuracy—at least for targets whose clinical definition is already explicitly morphological.

1.12.3 Vision–Language Models and Report Generation

The most recent direction outputs prose rather than labels. Islam Sk et al. present an explainable vision–language framework that classifies severity, segments the anatomy and generates radiologist-style reports, built on biomedical foundation models, with a spatial patch cross-attention module that preserves the anatomical hierarchies global pooling discards and an adaptive PID-Tversky loss for class imbalance; they report 90.69% classification accuracy, macro-averaged Dice 0.9512 and CIDEr 92.80% [10]. Zeng et al. take a narrower approach, fusing ConvNeXt image features with GPT-4o-generated descriptions for foraminal stenosis classification across three centres and seven scanners [66]. These systems are early and their language metrics measure fluency rather than clinical correctness, but they indicate where the field is heading.

1.12.4 Reader-Assistance Studies and Workflow Impact

The most consequential evidence for clinical value comes not from model metrics but from measuring what changes when readers use the model. Lim et al. conducted the definitive study of this kind: eight radiologists with 2–13 years of experience read the same studies with and without deep-learning assistance, separated by a one-month washout, against a 32-year-veteran musculoskeletal radiologist as reference. Interpretation time per study fell from 124–274 seconds to 47–71 seconds, and interobserver agreement was superior or equivalent for every stenosis grading. The largest gain fell to general and in-training radiologists on four-class foraminal stenosis, where κ rose from 0.39 to 0.71 and 0.70 [64]. Gao et al. observed a comparable effect for Modic assessment, where AI assistance raised agreement between a junior radiologist and a senior neuroradiologist from $\kappa = 0.52$ to 0.58 [21].

The pattern in both is that assistance helps least-experienced readers most and standardises rather than replaces judgement. This reframes the objective: the target is not a model that outperforms the best radiologist, but one that raises the floor.

1.12.5 Uncertainty and Calibration

Uncertainty estimation is conspicuously underdeveloped in this literature. Where errors concentrate at grade boundaries and where the reference standard is itself unreliable, a calibrated confidence would let borderline cases be referred rather than silently graded—a natural fit for triage. Almost none of the reviewed work reports calibration. Patil et al. are the exception, applying isotonic calibration on a validation split and reporting Brier scores alongside discrimination metrics [42]. Since modern networks are systematically overconfident [107], an uncalibrated severity probability should not be interpreted as a clinical confidence, and this remains an open opportunity rather than a solved problem.

1.13 Synthesis and Research Gap

1.13.1 What the Literature Has Established

Five findings are supported well enough to be treated as settled.

Anatomical localisation precedes classification. From DeepSPINE [62] through the interpretable pipelines of Bharadwaj et al. [14] to essentially every leading RSNA challenge solution [7], systems that localise the disc level before grading it outperform those that classify whole images. Detection is close to solved, with several studies reporting near-perfect level assignment [43, 68].

Automated grading is competitive with human reliability, because human reliability is low. Inter-reader agreement is substantial for central canal stenosis but only moderate for foraminal and poor for subarticular grading [25]. Against that benchmark, models sit inside the band of human disagreement [28, 35, 39]. This is a real result, but it is a statement about the modesty of the reference standard as much as the capability of the models.

Errors concentrate at adjacent grade boundaries. Disagreement above one grade is rare—0.85% of discs in the largest external validation [35]—and collapsing to a severe-versus-rest decision raises agreement substantially [40]. The difficulty lies in boundary placement, not gross misclassification.

Detection generalises across institutions; fine-grained grading does not. The asymmetry is consistent and large [34, 60, 94].

Class imbalance must be addressed explicitly. With 85.4% of spinal canal grades normal or mild [12], unweighted objectives yield degenerate solutions, and weighted sampling, weighted or focal losses and contrastive pretraining are all established mitigations [8, 95].

1.13.2 What Remains Contested

Three questions remain genuinely open, and the disagreement is substantive rather than terminological.

Does architectural sophistication help? The general computer vision literature would predict transformers to outperform convolutional networks given sufficient data, but the lumbar evidence does not bear this out. Kowlagi et al. report a tuned CNN pipeline beating transformer approaches [83]; Lee et al. find a CNN superior and a transformer merely comparable [91]; Silveira et al. beat TransUNet with an enhanced U-Net [75]; Udomluck et al. find conventional VGG19 features generalising better than DINOv2 [36]; and Patil et al. find deep features adding nothing over radiomics [42]. Set against this, hybrid architectures report gains [11, 78], and cross-sequence attention reports large ones [9].

Does the ordinal structure of the labels admit exploitation? Errors demonstrably cluster at adjacent grades, yet the most systematic test of ordinal-aware objectives found none improved on plain cross-entropy [41].

How much imaging is actually required? Clinical practice and the strongest benchmark results use all three sequences [7, 12], while sagittal-only systems report competitive performance [8, 83] and sagittal-only systems also fail in a specific, predicted way on the lateral compartments [29].

1.13.3 Gap 1: Architecture Comparison Under Controlled Conditions

The comparative studies above are informative but none isolates the variable of interest. Three confounds recur.

First, *unequal pipelines.* Compared architectures rarely receive identical preprocessing, localisation and training budgets, so a reported difference may reflect engineering effort rather than representational capacity—precisely the argument Kowlagi et al. advance when attributing their CNN's advantage to pipeline design rather than to convolution as such [83]. Given the evidence of Section 1.6.5 that localisation dominates performance, a comparison in which one arm is better localised than the other measures preprocessing.

Second, *frozen features.* Several comparisons extract features from frozen pretrained backbones rather than fine-tuning end to end [36, 42]. This systematically disadvantages transformers, whose weaker inductive biases make them more dependent on in-domain adaptation.

Third, *fixed input configuration.* Every comparison holds the sequence configuration constant, so the interaction between architecture and available context is untested. This matters because the theoretical case for global attention is strongest exactly where context must be inferred rather than observed—that is, where a view is missing.

A controlled comparison therefore requires a shared localisation stage, a shared training and augmentation protocol, comparable capacity and tuning budgets, and end-to-end fine-tuning of both backbones, evaluated on the same benchmark with the same metric. No study in this review satisfies all four conditions.

1.13.4 Gap 2: Sequence Configuration and the Accuracy–Efficiency Trade-off

The second gap concerns input rather than model. Multi-sequence pipelines carry costs that are rarely quantified alongside their accuracy: greater memory and compute, cross-plane geometric registration dependent on DICOM metadata [7, 68], longer inference, and brittleness when a sequence is missing or degraded (Section 1.8.5). Against these, the incremental diagnostic yield of each additional sequence has not been measured under controlled conditions.

The literature supplies both sides of the argument but no direct test. Evidence that the third sequence matters is indirect, inferred from the failure of sagittal-only systems on foraminal and subarticular targets [29, 94]; evidence that it may not is likewise indirect, resting on sagittal-only systems that perform respectably against different baselines on different data [8, 83]. Because these come from different architectures, cohorts, localisation strategies and metrics, the contribution of sequence configuration cannot be separated from everything else that differs.

Two design requirements follow. The comparison must hold architecture and pipeline constant while varying only the input configuration; and it must report results per condition rather than in aggregate, since the expected penalty is concentrated in the foraminal and subarticular targets and an aggregate figure would average away the structure of interest.

1.13.5 Research Questions and Objectives

The two gaps combine into a single factorial investigation. Crossing architecture with input configuration answers both questions and, additionally, tests whether they interact—whether a transformer's global attention can partially compensate for absent multiplanar data, which neither gap addresses alone. This yields the following research questions:

RQ1 Under matched localisation, training and evaluation protocols, how do convolutional backbones (ResNet50, EfficientNet-B4) compare with a hierarchical vision transformer (Swin) for per-level, per-condition severity grading of lumbar degenerative disease?

RQ2 What is the cost in diagnostic performance of reducing the input from the full multi-sequence configuration (Sagittal T1, Sagittal T2/STIR, Axial T2) to a single-sequence configuration (Sagittal T2), and how is that cost distributed across the five graded conditions?

RQ3 Do architecture and sequence configuration interact—specifically, is the penalty for single-sequence input smaller for the transformer than for the convolutional backbones?

RQ4 Which class-imbalance strategies most effectively reduce clinically consequential errors, in particular severe cases graded as normal or mild?

The corresponding objectives are to implement a shared localisation and preprocessing stage so that all arms receive identical input regions; to train each architecture under both configurations with matched budgets; to evaluate using the benchmark's weighted metric together with per-condition and per-class reporting; and to quantify computational cost—parameters, memory and inference time—so that any accuracy difference can be weighed against it.

1.13.6 Scope and Delimitations

Four boundaries are stated explicitly. The work addresses *severity grading*, taking localisation as a shared and largely solved precondition (Section 1.6.1) rather than a contribution. It evaluates on *LumbarDISC* [12]; because that collection excludes studies with hardware, non-degenerative pathology, severe scoliosis or substantial artifact, conclusions do not extend to those populations, and no independent external cohort is available to test transportability beyond it. It compares a *bounded set of architectures*—two convolutional and one transformer—chosen as representative rather than exhaustive, and does not evaluate hybrid or vision–language designs [10, 78]. Finally, it reports model performance and computational cost, not clinical impact: the reader-assistance evidence of Section 1.12.4 [64] shows that workflow benefit requires a prospective reader study, which lies outside the present scope.

Two constraints inherited from the literature bound the interpretation of any result. Performance is capped by the reliability of the reference standard (Section 1.4.7), so figures approaching perfect agreement should be read as evidence of a protocol artefact rather than of clinical readiness. And single-benchmark evaluation cannot establish generalisation; the consistent development-to-validation drop documented in Section 1.11.3 applies to this work as much as to any other.

1.14 Chapter Summary

This chapter reviewed the literature on automated assessment of lumbar degenerative disease from MRI, from the clinical grading systems that define the task through to the architectures that currently attempt it.

The clinical sections established that severity labels originate in morphological grading systems—Pfirrmann for disc degeneration [18], Schizas and Lee for central canal [13, 23], Lee for foraminal [24]—whose inter-reader reliability ranges from substantial to poor depending on the compartment [25]. This imposes a ceiling on achievable performance and makes agreement statistics, rather than accuracy, the appropriate measure. The labels are ordinal, errors concentrate at adjacent boundaries, and the clinical cost of an error depends on its direction.

The methodological sections traced the field from handcrafted features through SpineNet's multi-task convolutional grading [50] to contemporary multi-stage, multi-view pipelines [7]. The most robust design principle to emerge is that anatomical localisation should precede classification [62, 83]. The most robust empirical regularity is that detection transfers across institutions while fine-grained grading does not [34, 94].

Two questions remain unresolved. Whether attention-based backbones offer any advantage over well-tuned convolutional ones is contested, with the balance of lumbar-specific evidence currently favouring CNNs [75, 83, 91], yet no study has compared them under matched localisation, training and evaluation conditions. And whether the full three-sequence protocol is necessary, or whether a single ubiquitous sequence suffices, has been argued from both sides without a controlled test. These two questions, and their interaction, constitute the gap this thesis addresses. Chapter 3 sets out the experimental design through which they are answered.

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